

PROJECT REPORT

Auto Ticket Classification using Flow Designer

ServiceNow IT Helpdesk Automation Project

Team Members

Team Member	Role
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1. INTRODUCTION

1.1 Project Overview

The project, “Auto Ticket Classification using Flow Designer,” is a ServiceNow-based automation solution for a school IT helpdesk. The system handles common support incidents such as Wi-Fi problems, projector issues, password/login problems, and slow or hanging computers. Instead of requiring each ticket to be categorized manually, the system uses predefined keyword conditions in ServiceNow Flow Designer to automatically assign the appropriate Category and Subcategory.

A custom Incident WorkFlow table is used to store ticket information including Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned To. The automation is designed as a no-code/low-code workflow that can be maintained by updating conditions and branches. [filecite?turn0file0?L3-L11? filecite?turn0file0?L78-L91?](#)

1.2 Purpose

The purpose of the project is to reduce manual ticket classification, improve consistency in categorization and routing, and provide a maintainable automation approach for school IT support. The project automatically classifies a ticket at creation and assigns the relevant category and subcategory based on the issue description. [filecite?turn0file0?L28-L47?](#)

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives incidents related to network connectivity, classroom projectors, password or login access, and computer performance. When tickets are manually categorized, support staff must repeatedly inspect the issue description and select the correct category and subcategory. This creates additional manual effort and can lead to inconsistent classification or delayed routing.

The proposed project addresses this problem by automatically identifying predefined keywords in the ticket description and assigning the corresponding Category and Subcategory through ServiceNow Flow Designer. [filecite@turn0file0L3-L11](#)

2.2 Empathy Map Canvas

The primary user considered is the school IT helpdesk/support staff member who receives and processes incidents.

Empathy Map Area	Project-Specific Content
Says	“I receive many similar IT support tickets.” “I need tickets categorized correctly.”
Thinks	Manual classification takes time and repeated selection may cause mistakes.
Does	Reviews ticket descriptions, selects categories, routes incidents, and communicates with users.
Feels	Needs a faster and more consistent way to process incoming incidents.
Pains	Manual categorization, repetitive work, incorrect routing, and delayed handling.
Gains	Automatic classification, consistent categories, reduced effort, and improved routing.

2.3 Brainstorming

The brainstorming phase focused on practical ways to reduce manual classification. The selected idea was to use ServiceNow Flow Designer because it supports trigger-based, condition-based, no-code/low-code automation.

- Detect a newly created ticket automatically.
- Check whether the Category field is empty.
- Search the ticket description for supported keywords.
- Assign Category and Subcategory according to the matching condition.
- Send an email notification to the caller after processing.
- Keep the solution maintainable by allowing new keyword branches to be added.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Activity	Problem	System Response
Ticket Creation	User/support staff creates an IT ticket.	Ticket is initially uncategorized.	Flow is triggered when the record is created.
Issue Description	User enters a short description/description.	Issue must be interpreted manually.	Flow checks predefined keywords.
Classification	Support staff normally selects category/subcategory.	Manual selection consumes time.	Flow assigns the matching Category/Subcategory.

Routing	Ticket is prepared for support handling.	Incorrect category can affect routing.	Structured category information is stored.
Notification	Caller waits for confirmation.	Manual communication may be required.	Email notification action is executed.

3.2 Solution Requirement

The main requirements are automatic classification at ticket creation, Category/Subcategory assignment, dependent-choice logic, email notification, structured storage, and a scalable configuration that can be extended with additional classification rules. [filecite?turn0file0?L28-L47?](#)

Requirement	Description
Automatic Classification	Classify newly created tickets using predefined keywords.
Category Assignment	Assign Network, Hardware, Access, or Performance.
Subcategory Assignment	Assign Wi-Fi, Projector, Forgot Password, or Slow Computer.
Dependent Choice	Maintain valid Category/Subcategory relationships.
Notification	Notify the caller after ticket processing.
Maintainability	Allow classification conditions to be modified in Flow Designer.

3.3 Data Flow Diagram

The logical data flow is: Caller → Incident WorkFlow Ticket → Flow Designer Trigger → Keyword/Condition Evaluation → Category & Subcategory Update → Email Notification → Classified Ticket.

A detailed diagram can be inserted in this section as the project's Data Flow Diagram screenshot.

3.4 Technology Stack

Technology	Purpose
ServiceNow	Application and IT service management platform.
Flow Designer	Implements the ticket classification automation.
Incident WorkFlow Table	Stores ticket and assignment information.
Dependent Choice Logic	Maintains valid Category/Subcategory combinations.
Email Action	Sends notification to the caller.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The solution directly addresses the repetitive manual task identified in the problem statement. Since the supported ticket types can be recognized through predefined keywords, a rule-based Flow

Designer implementation can classify these common incidents without requiring a machine-learning model.

Problem	Proposed Fit
Manual categorization	Automatic Category/Subcategory assignment.
Repeated inspection of descriptions	Keyword-based condition branches.
Incorrect category combinations	Dependent-choice logic.
Manual user notification	Automated email action.
Difficulty maintaining rules	Flow Designer branches can be modified without traditional application code.

4.2 Proposed Solution

When a new Incident WorkFlow record is created, the flow checks whether the Category is empty. It then evaluates the configured conditions. Wi-Fi/WiFi maps to Network → Wi-Fi; Projector maps to Hardware → Projector; Password/Login maps to Access → Forgot Password; and Slow/Hanging maps to Performance → Slow Computer. After classification, the workflow can send an email notification to the caller. [filecite?turn0file0?L202-L220?filecite?turn0file0?L222-L250?](#)

4.3 Solution Architecture

The architecture consists of the ServiceNow ticket interface, the Incident WorkFlow data layer, and the Flow Designer automation layer. A new record enters the workflow, the trigger starts the flow, conditions identify the ticket type, and the corresponding fields are updated. The notification action completes the workflow.

Layer	Component	Responsibility
User Layer	ServiceNow Ticket Form	Enter and view IT incident details.
Data Layer	Incident WorkFlow	Store ticket, category, subcategory, state, and assignment fields.
Automation Layer	Flow Designer	Trigger, evaluate conditions, update classification, and notify.
Validation Layer	Testing/UAT	Verify expected classifications and workflow behavior.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned as a short three-sprint implementation from 28 September 2026 to 3 October 2026. The work was divided into requirements/configuration, automation and testing, and final validation/documentation.

Sprint	Story Points	Duration	Dates	Major Activities
Sprint 1	19	2 Days	28–29 Sep 2026	Requirements, table/field setup,

				category and subcategory configuration.
Sprint 2	26	2 Days	30 Sep–1 Oct 2026	Flow Designer automation, keyword branches, notification configuration.
Sprint 3	17	2 Days	2–3 Oct 2026	Testing, UAT, documentation, final validation.
Total	62	6 Days	28 Sep–3 Oct 2026	Complete project implementation.

5.2 Sprint Planning

Sprint 1 focused on establishing the data structure and requirements. Sprint 2 implemented the automation and notification workflow. Sprint 3 focused on functional/performance testing, UAT, documentation, and final validation.

5.3 Sprint Burndown Chart

For Sprint 1, the planned story points were 19. The burndown plan was Day 0 = 19 points, Day 1 = 9 points, and Day 2 = 0 points. Insert the Sprint 1 Burndown Chart screenshot below this section.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test ID	Scenario	Expected Result	Result
FT-01	Ticket creation	Ticket is created successfully.	Pass
FT-02	Wi-Fi keyword	Network → Wi-Fi.	Pass
FT-03	Projector keyword	Hardware → Projector.	Pass
FT-04	Password/Login keyword	Access → Forgot Password.	Pass
FT-05	Slow/Hanging keyword	Performance → Slow Computer.	Pass
FT-06	Email notification	Notification is sent to caller.	Pass

6.2 Performance Testing

Performance testing verifies that the Flow Designer classification executes automatically without a noticeable delay, that multiple tickets can be processed independently, and that the resulting Category, Subcategory, State, Assigned Group, and Assigned To values remain valid and consistent.

6.3 Model Performance Testing

Traditional Training Accuracy and Validation Accuracy are not applicable because the project does not train a machine-learning model. Functional classification accuracy can instead be calculated as: $\text{Correctly Classified Tickets} \div \text{Total Test Tickets} \times 100$. Confidence score is also not applicable because the project does not use YOLO or object detection.

6.4 User Acceptance Testing (UAT)

UAT covers ticket creation, the four supported classification scenarios, email notification, multiple-ticket processing, data validation, and handling of an unsupported/unknown keyword. The documented UAT scenarios are intended to confirm that the automation behaves as expected from the user's perspective.

7. RESULTS

7.1 Ticket Classification Results

Input / Keyword	Category	Subcategory	Result
Wi-Fi / WiFi	Network	Wi-Fi	Pass
Projector	Hardware	Projector	Pass
Password / Login	Access	Forgot Password	Pass
Slow / Hanging	Performance	Slow Computer	Pass

7.2 Test Results

The documented functional scenarios passed successfully. The project uses rule-based classification, so the main validation criterion is whether each supported ticket description produces the correct Category and Subcategory. The testing also verifies notification behavior and data validity.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces repetitive manual ticket classification.
- Provides consistent Category and Subcategory assignment for supported keywords.
- Improves the structure of helpdesk ticket data.
- Uses ServiceNow Flow Designer, reducing the need for traditional application code.
- Can be extended by adding new conditions and classification branches.
- Supports automated caller notification.

Disadvantages

- Classification depends on predefined keywords and conditions.
- Unsupported wording may not be classified automatically.
- The solution does not learn automatically from new ticket examples.
- Complex tickets containing multiple unrelated issues may require additional rules or manual review.
- The automation depends on correct configuration of the ServiceNow fields and flow.

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project provides an end-to-end ServiceNow automation for common school IT helpdesk incidents. The implementation uses a custom Incident WorkFlow table, structured fields, dependent Category/Subcategory choices, and Flow Designer conditions to classify tickets automatically. The documented testing approach verifies ticket creation, classification, notification, data validation, and multiple-ticket processing. The solution provides a maintainable foundation that can be extended with additional classification rules and support scenarios. [filecite?turn0file0?L454-L459?](#)

10. FUTURE SCOPE

- Expand the keyword library to recognize more variations of helpdesk issues.
- Add more IT service categories and subcategories.
- Add advanced routing to automatically assign tickets to specific support groups.
- Add escalation rules for high-priority or unresolved incidents.
- Introduce dashboards and reports if ticket analytics become a project requirement.
- Improve handling of multiple keywords and multi-issue tickets.
- Explore AI/ML-assisted classification as a future enhancement if the project scope expands beyond rule-based automation.

11. APPENDIX

11.1 ServiceNow Configuration Screenshots

Insert screenshots of the Incident WorkFlow table, ticket form, fields, and Category/Subcategory configuration.

11.2 Flow Designer Screenshots

Insert screenshots showing the Record Created trigger, Category-empty condition, keyword branches, field updates, and email action.

11.3 Test Case Results

Insert screenshots of the functional testing and UAT results.

11.4 Project Documentation

Include the completed project documentation, testing documents, UAT report, sprint planning content, and related project artifacts.

11.5 Project Demo Link

GitHub Link: <https://github.com/Abinaya959/Auto-Ticket-Classification-using-Flow-Designer/tree/main>